
SI 001 PSU - Sensor Interface

Installation and operating instructions

(GB) (D) (F) (I) (E) (GR) (DK) (PL) (RU) (H) (CZ)



(GB) Declaration of Conformity

We, Grundfos, declare under our sole responsibility that the product SI 001 PSU - Sensor Interface, to which this declaration relates, is in conformity with these Council directives on the approximation of the laws of the EC member states:

- Low Voltage Directive (2006/95/EC).
Standard used: EN 61010-1: 2001
- EMC Directive (2004/108/EC).
Standard used: EN 61326-1: 2006

(F) Déclaration de Conformité

Nous, Grundfos, déclarons sous notre seule responsabilité, que le produit SI 001 PSU - Sensor Interface, auquel se réfère cette déclaration, est conforme aux Directives du Conseil concernant le rapprochement des législations des Etats membres CE relatives aux normes énoncées ci-dessous :

- Directive Basse Tension (2006/95/CE).
Norme utilisée : EN 61010-1: 2001
- Directive Compatibilité Electromagnétique CEM (2004/108/CE).
Norme utilisée : EN 61326-1: 2006

(E) Declaración de Conformidad

Nosotros, Grundfos, declaramos bajo nuestra propia responsabilidad que el producto SI 001 PSU - Sensor Interface, al cual se refiere esta declaración, está conforme con las Directivas del Consejo en la aproximación de las leyes de los Estados Miembros del EM:

- Directiva de Baja Tensión (2006/95/CE).
Norma aplicada: EN 61010-1: 2001
- Directiva EMC (2004/108/CE).
Norma aplicada: EN 61326-1: 2006

(DK) Overensstemmelseserklæring

Vi, Grundfos, erklærer under ansvar at produktet SI 001 PSU - Sensor Interface som denne erklæring omhandler, er i overensstemmelse med disse af Rådets direktiver om indbyrdes tilnærmelse til EF-medlemsstaternes lovgivning:

- Lavspændingsdirektivet (2006/95/EF).
Anvendt standard: EN 61010-1: 2001
- EMC-direktivet (2004/108/EF).
Anvendt standard: EN 61326-1: 2006

(RU) Декларация о соответствии

Мы, компания Grundfos, со всей ответственностью заявляем, что изделия SI 001 PSU - Sensor Interface, к которым относится настоящая декларация, соответствуют следующим Директивам Совета Евросоюза об унификации законодательных предписаний стран-членов ЕС:

- Низковольтное оборудование (2006/95/EC).
Применявшийся стандарт: EN 61010-1: 2001
- Электромагнитная совместимость (2004/108/EC).
Применявшийся стандарт: EN 61326-1: 2006

(CZ) Prohlášení o shodě

My firma Grundfos prohlašujeme na svou plnou odpovědnost, že výrobek SI 001 PSU - Sensor Interface, na nějž se toto prohlášení vztahuje, je v souladu s ustanoveními směrnice Rady pro sblížení právních předpisů členských států Evropského společenství v oblastech:

- Směrnice pro nízkonapěťové aplikace (2006/95/ES).
Použitá norma: EN 61010-1: 2001
- Směrnice pro elektromagnetickou kompatibilitu (EMC) (2004/108/ES).
Použitá norma: EN 61326-1: 2006

(D) Konformitätserklärung

Wir, Grundfos, erklären in alleiniger Verantwortung, dass das Produkt SI 001 PSU - Sensor Interface, auf das sich diese Erklärung bezieht, mit den folgenden Richtlinien des Rates zur Angleichung der Rechtsvorschriften der EU-Mitgliedsstaaten übereinstimmt:

- Niederspannungsrichtlinie (2006/95/EG).
Norm, die verwendet wurde: EN 61010-1: 2001
- EMV-Richtlinie (2004/108/EG).
Norm, die verwendet wurde: EN 61326-1: 2006

(I) Dichiarazione di Conformità

Grundfos dichiara sotto la sua esclusiva responsabilità che il prodotto SI 001 PSU - Sensor Interface, al quale si riferisce questa dichiarazione, è conforme alle seguenti direttive del Consiglio riguardanti il riavvicinamento delle legislazioni degli Stati membri CE:

- Direttiva Bassa Tensione (2006/95/CE).
Norma applicata: EN 61010-1: 2001
- Direttiva EMC (2004/108/CE).
Norma applicata: EN 61326-1: 2006

(GR) Δήλωση Συμμόρφωσης

Εμείς, η Grundfos, δηλώνουμε με αποκλειστικά δική μας ευθύνη ότι τα προϊόντα SI 001 PSU - Sensor Interface, στα οποία αναφέρεται η παρούσα δήλωση, συμμορφώνονται με τις εξής Οδηγίες του Συμβουλίου περί προσγγίσις των νομοθεσιών των κρατών μελών της ΕΕ:

- Οδηγία χαμηλής τάσης (2006/95/ΕC).
Πρότυπο που χρησιμοποιήθηκε: EN 61010-1: 2001
- Οδηγία Ηλεκτρομαγνητικής Συμβατότητας (EMC) (2004/108/ΕC).
Πρότυπο που χρησιμοποιήθηκε: EN 61326-1: 2006

(PL) Deklaracja zgodności

My, Grundfos, oświadczamy z pełną odpowiedzialnością, że nasze wyroby SI 001 PSU - Sensor Interface, których deklaracja niniejsza dotyczy, są zgodne z następującymi wytycznymi Rady d/s ujednolicenia przepisów prawnych krajów członkowskich WE:

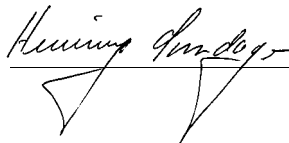
- Dyrektywa Niskonapięciowa (LVD) (2006/95/WE).
Zastosowana norma: EN 61010-1: 2001
- Dyrektywa EMC (2004/108/WE).
Zastosowana norma: EN 61326-1: 2006

(H) Megfelelősségi nyilatkozat

Mi, a Grundfos, egyedüli felelősséggel kijelentjük, hogy a SI 001 PSU - Sensor Interface termék, amelyre jelen nyilatkozik vonatkozik, megfelel az Európai Unió tagállamainak jogi irányelveit összehangoló tanács alábbi előírásainak:

- Kisfeszültségű Direktíva (2006/95/EK).
Alkalmazott szabvány: EN 61010-1: 2001
- EMC Direktíva (2004/108/EK).
Alkalmazott szabvány: EN 61326-1: 2006

Bjerringbro, 1st July 2009



Henning Sandager
Technical Director

SI 001 PSU - Sensor Interface

Installation and operating instructions

GB

Montage- und Betriebsanleitung

D

Notice d'installation et d'entretien

F

Istruzioni di installazione e funzionamento

I

Instrucciones de instalación y funcionamiento

E

Οδηγίες εγκατάστασης και λειτουργίας

GR

Monterings- og driftsinstruktion

DK

Instrukcja montażu i eksploatacji

PL

Руководство по монтажу и эксплуатации

RU

Szerelési és üzemeltetési utasítás

H

Montážní a provozní návod

CZ

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Warning
Prior to installation, read these installation and operating instructions. Installation and operation must comply with local regulations and accepted codes of good practice.

1. Symbols used in this document



Warning
If these safety instructions are not observed, it may result in personal injury!



Caution
If these safety instructions are not observed, it may result in malfunction or damage to the equipment!



Note
Notes or instructions that make the job easier and ensure safe operation.

2. Applications

The Grundfos SI sensor interface is used as external power supply for the DPI differential pressure transmitter.

2.1 Nameplate

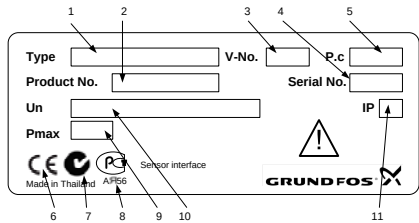


Fig. 1 Example of nameplate

Pos.	Description
1	Type designation
2	Product number
3	Version number
4	Serial number
5	Production code (year and week)
6, 7, 8	Approvals and CE marking
9	Maximum power consumption
10	Supply voltage
11	Enclosure class

2.2 Explanation to symbols

Symbol	Designation	Description
	LED1	Red/green status LED for main network. Not used in this model.
	LED2	Red/green status LED for internal communication between the interface and the Grundfos product. Not used in this model.

TM04 4132 0809

3. Installation kit

The SI package contains these parts:

- interface with six screwed cable entries with blanking plugs
- DIN rail clip with screws
- installation and operating instructions
- Quick Guide.

4. Installation

The interface unit can be mounted on a wall or on a DIN rail in a control cabinet.

4.1 Wall mounting

Mount the interface on the wall. See fig. 2 and the drilling template in the Quick Guide to the interface.

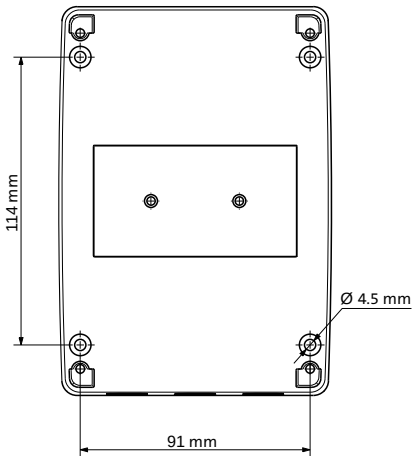


Fig. 2 Wall mounting of the interface

4.2 Mounting in control cabinet

When the interface is mounted in a control cabinet containing a frequency converter, it is especially important to ensure an EMC-correct installation of both the interface and frequency converter.

The following rules must therefore be observed:

- All cable screens must be connected to earth. If it is not possible to use cable clamps, the stripped part of the cable screen must be as short as possible to reduce the impedance at high frequencies.
- All units in the control cabinet must be connected to earth.

Mounting on DIN rail

The interface is prepared for mounting on a 35 mm DIN rail (EN 50022).

1. Fix the DIN rail clip supplied on the interface. See the instructions on the interface.
2. Fit the interface by hooking the top on the DIN rail and clicking the bottom onto the rail. See fig. 3.



Fig. 3 Sensor interface on DIN rail

Dismounting from DIN rail

Remove the interface by pulling it downwards and free of the DIN rail.

4.3 Electrical connection

The electrical connection should be carried out by an authorised electrician in accordance with local regulations.

Warning



Before beginning the electrical connection, make sure that the electricity supply has been switched off and that it cannot be accidentally switched on.

Caution

Supply voltage to the interface: 115-400 VAC 50-60 Hz, ± 10 %.

Note

If the supply cable to the interface has no earth conductor, a separate earth connection must be connected to the internal earth terminal. See fig. 4, pos. 4.

TM04 1727 1008

TM04 1724 1008

4.3.1 Connection

- 1. Remove the cover.
- 2. Connect the supply cable to the interface (fig. 4, pos. 1, 2 and 3).
- 3. If the supply cable has no earth conductor, connect an external earth connection (pos. 4, fig. 4). Loosen the earth clamp before connecting the supply cable/earth conductor.
- 4. Connect the transmitter and controller cables to the interface. See fig. 5. Cut the white wire of the DPI. Connect the cable screen of both the transmitter and controller cable under the earth clamp.
- 5. Fit the cover.

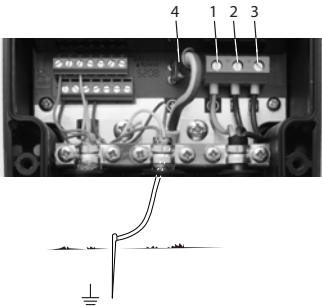


Fig. 4 Electrical connection and earthing

Pos.	Designation	Description
1	J3-1	Earth terminal
2	J3-2	Phase terminal (L1)
3	J3-3	Neutral terminal or (L2)
4	J2-1	Internal earth terminal

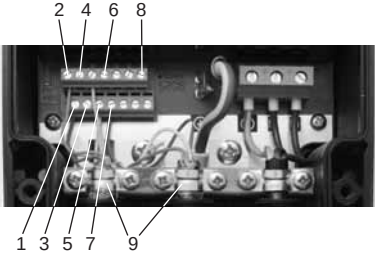


Fig. 5 Sensor and controller connection

Pos.	Designation	Description
1	J1-1	Sensor 1, transmitter signal (green DPI conductor)
2	J1-2	Sensor 1, controller signal
3	J1-3	Sensor 2, transmitter signal (green DPI conductor)
4	J1-4	Sensor 2, controller signal
5	J1-5	GND (yellow DPI conductor)
6	J1-6	Controller signal 1
7	J1-7	+24 V (brown DPI conductor)
8	J1-14	Controller signal 2
9	-	Screen from transmitter and controller

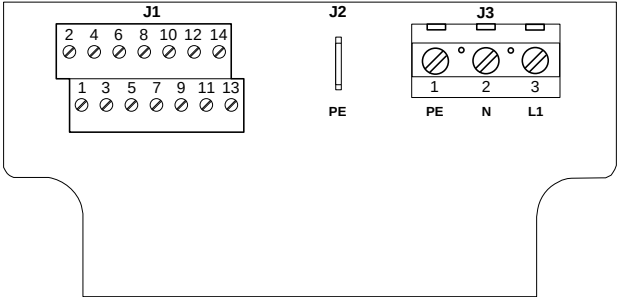


Fig. 6 Connecting terminals

TM04 4195 1009

TM04 4196 1009

TM04 4131 0809

5. Technical data

Supply voltage				
Supply voltage		115-400 VAC ±10 %		
Frequency		50-60 Hz		
Power consumption		Maximum 7 W		
Cable size		IEC: 0.75 - 4 mm ² UL: 19-12 AWG		
Transmitter cable				
Recommended cable type		Screened Cross-section: 0.25 - 1 mm ² AWG: 24-18 Maximum cable length: 30 m/100 ft		
Controller				
Controller connection		2-wire, 4-20 mA		
Recommended cable type		Screened, twisted-pair Cross-section: 0.25 - 1 mm ² AWG: 23-18		
Supply cable				
Conductor size			Maximum cable length [m]	
AWG	Diameter [mm]	Cross-sectional area [mm ²]	Resistance load 500 Ω	Resistance load 250 Ω
17	1.15	1.04	4834	10877
18	1.02	0.82	3040	6840
19	0.91	0.65	1912	4302
20	0.81	0.52	1202	2706
21	0.72	0.41	756	1702
22	0.64	0.33	476	1070
23	0.57	0.26	299	673
Other data				
Enclosure class		IP54		
Cable entry		6 x M16 Ø4 - Ø10		
Ambient temperature				
During operation		−20 °C to +50 °C (−4 °F to +122 °F)		
During storage		−20 °C to +70 °C (−4 °F to +158 °F)		
During transportation		−20 °C to +70 °C (−4 °F to +158 °F)		

Caution

During operation, the interface must not be exposed to direct sunlight.

6. Disposal

This product or parts of it must be disposed of in an environmentally sound way:

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.

Subject to alterations.

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